

2.1 Equations of Motion

Question Paper

Course	CIE A Level Physics
Section	2. Kinematics
Topic	2.1 Equations of Motion
Difficulty	Hard

Time allowed: 60

Score: /44

Percentage: /100

Question 1a

During a fireworks display, a firework is launched into the air and explodes once it reaches a maximum height of 250 m.

The vertical component of the velocity at launch depends on both the initial velocity of the firework and the angle θ between the initial velocity and the horizontal.

The maximum initial velocity a firework can be launched at is 110 m s^{-1} .

(i)

Calculate the initial launch velocity for a firework that is fired directly upwards.

[2]

(ii)

Show that the minimum launch angle between the initial velocity and the horizontal to reach 250 m is about 40° .

[3]

[5 marks]

Question 1b

On Fig. 1.1, sketch the variation of angle θ with the initial velocity required for the firework to reach the maximum height of 250 m.

Include values from (a) on the axes.

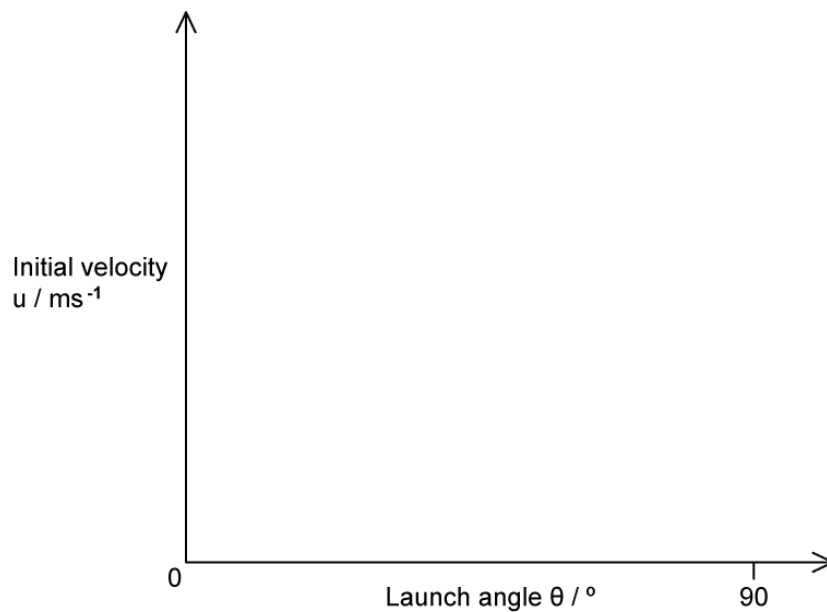


Fig. 1.1

[4 marks]

Question 1c

The length of the fuse within the firework is chosen to ensure it explodes just as it reaches its highest point above the ground.

The firework is launched into the air at an angle of 75° to the horizontal, as shown in Fig. 1.2.

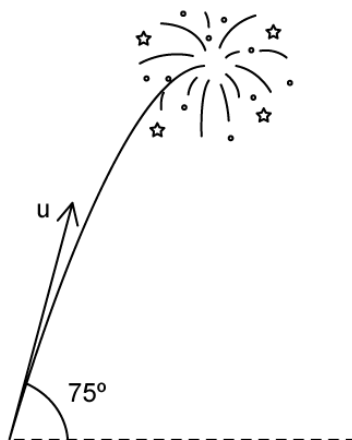


Fig. 1.2

The firework contains a fuse with a burn rate of 1.5 cm s^{-1} .

Determine the length of the fuse used in the firework.

[4 marks]

Question 1d

During the display, one of the fireworks is lit but falls over before launching. When it launches, it leaves the ground with an initial velocity of 75 m s^{-1} at an angle of 30° to the horizontal, as shown in Fig. 1.3.

The firework moves in the direction of a hill. The side of the hill is inclined at 8.5° to the horizontal, as shown.

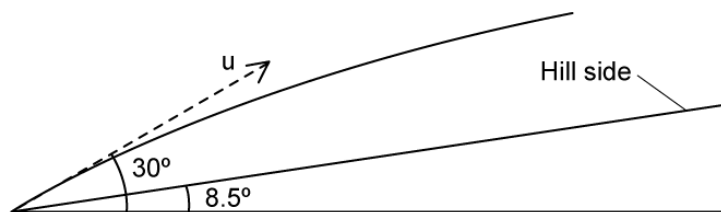


Fig. 1.3

Deduce whether the firework ignites before it lands on the hillside.

[4 marks]

Question 2a

Fig. 1.1 shows a stunt motorcyclist descending down a ramp.

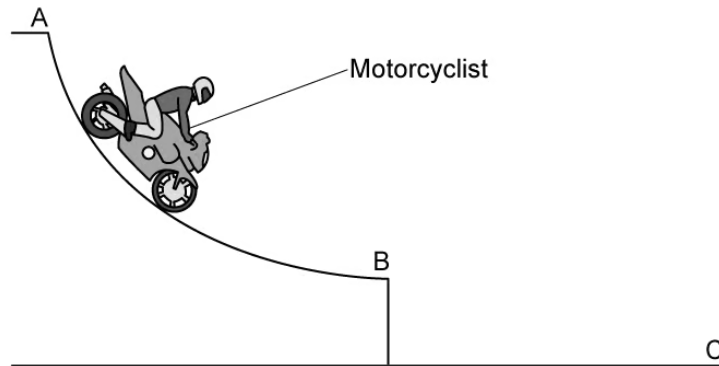


Fig. 1.1

The motorcyclist starts from rest at the top of the ramp at A and leaves the ramp at B horizontally.

After leaving the ramp at B, the motorcyclist lands on the ground at C, which is 6.4 m away. At C, the motorcyclist lands with a resultant velocity of 13.4 m s^{-1} at an angle of 29.5° with the horizontal.

Calculate the initial velocity of the motorcyclist at B.

In this question, assume that air resistance is negligible.

[4 marks]

Question 2b

The motorcyclist practices a stunt on a different ramp inclined at 25° and 3 m high as shown in Fig. 1.2.

The other ramp is 24 m away and has a height of 1 m, as shown.

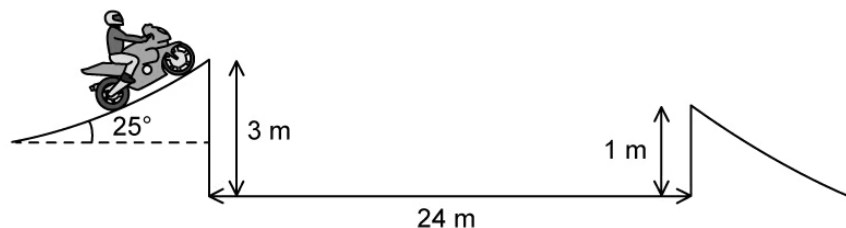


Fig. 1.2

Calculate the minimum initial velocity the motorcyclist requires in order to reach the other ramp.

[4 marks]

Question 2c

The stunt motorcyclist's ultimate goal is to jump across a river and land on the other side. Fig. 1.3 shows the motorcyclist driving off a ramp at the edge of a river.

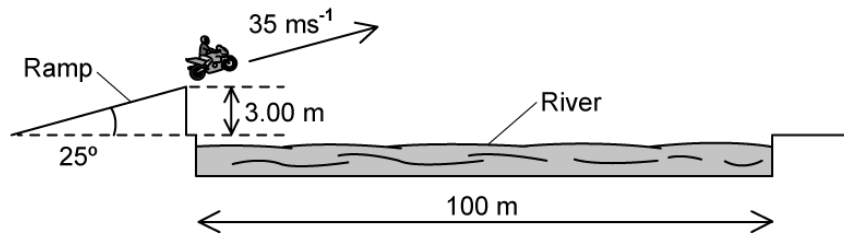


Fig. 1.3

The ramp is at an angle of 25° to the horizontal and the height at the end of the ramp is 3.0 m. The width of the river is 100 m. The initial velocity of the motorcyclist is 35 m s^{-1} .

Deduce whether the motorcyclist lands on the other side of the river.

[4 marks]

Question 2d

Explain how air resistance would affect the jump.

[3 marks]

Question 3a

An object is released near the surface of the Moon at time $t = 0$. Fig. 1.1 shows the variation of displacement s with time t of the object from the point of release.

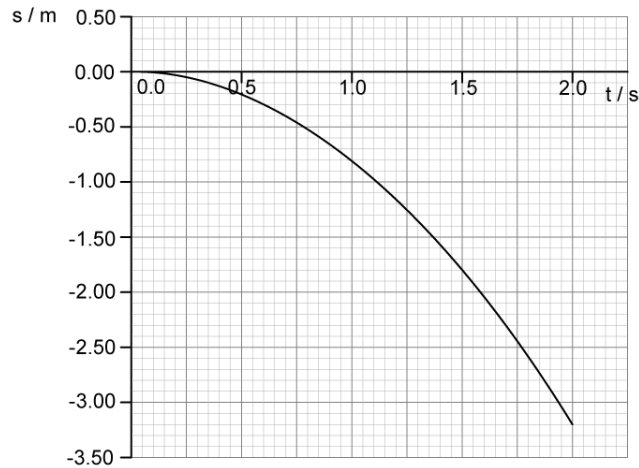


Fig. 1.1

- (i)
State the significance of the negative values of s .

[1]

- (ii)
State two assumptions about the motion of the object.

[2]

[3 marks]

Question 3b

Use the graph in Fig. 1.1 to determine a value for the acceleration of free fall close to the surface of the Moon.

[2 marks]

Question 3c

Use Fig. 1.1 to estimate the instantaneous velocity of the object when $t = 1.5$ s.

[3 marks]**Question 3d**

(i)

On Fig. 1.1, sketch the variation of displacement s with time t if the same object was released close to the surface of the Earth instead.

[2]

(ii)

Describe and explain the features of your sketch.

[2]**[4 marks]**

Model Answers: Hard

1a

(a)

List the known quantities:

- Maximum height, $s = 250 \text{ m}$
- Acceleration, $a = g = -9.81 \text{ m s}^{-2}$
- Initial velocity = u
- Final velocity, $v = 0$

(i) Calculate the initial launch velocity for a firework that is fired directly upwards:

- $v^2 = u^2 + 2as$
- $0 = u^2 + (2 \times -9.81 \times 250) = u^2 + -(2 \times 9.81 \times 250)$ [1 mark]
- $u = \sqrt{2 \times 9.81 \times 250}$
- $u = 70.0 \text{ m s}^{-1}$ [1 mark]

(ii) Determine the minimum launch angle:

- Maximum launch velocity, $u_{\max} = 110 \text{ m s}^{-1}$

Determine the vertical component at an angle θ to the horizontal:

- Vertical component of velocity: $u_v = u \sin \theta$ [1 mark]

Determine the minimum angle required to reach 250 m:

- $v^2 = u^2 + 2as \Rightarrow v^2 = u^2 - 2as$ since a is negative
- $0 = (u_{\max} \sin \theta_{\min})^2 - 2as \Rightarrow \sin \theta_{\min} = \frac{\sqrt{2as}}{u_{\max}}$

- $\theta_{min} = \sin^{-1} \left(\frac{\sqrt{2 \times 9.81 \times 250}}{110} \right)$ [1 mark]
- $\theta_{min} = 39.55 = 40^\circ$ [1 mark]

[Total: 5 marks]

The sign convention is **very** important for this question.

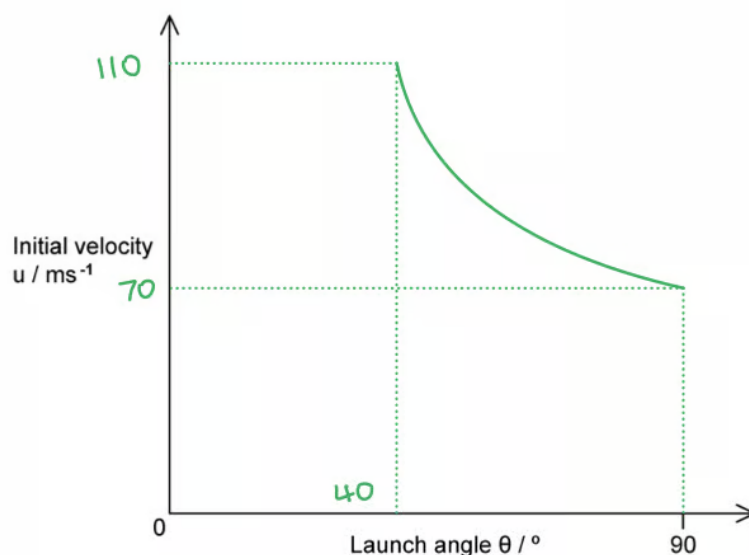
Here, the upwards direction has been taken as positive. So s and $u \sin \theta$ are positive but g always acts downwards, so is negative. You can take the downward direction as positive again, as long as you keep g positive and s and $u \sin \theta$ negative. Either convention will get your full marks as you'll get the same final answer.

1b

(b) The variation of angle θ with the initial velocity required is as follows:

List the known quantities:

- Minimum launch angle, $\theta = 40^\circ$
- Minimum initial velocity = 110.0 m s^{-1}
- Maximum initial velocity = 70 m s^{-1} (from part a)



- Curve drawn with negative gradient i.e. θ increases as u decreases; [1 mark]
- Minimum initial velocity marked, i.e. when $\theta_{max} = 90^\circ$, $u_{min} = 70 \text{ m s}^{-1}$; [1 mark]
- Minimum launch angle marked, i.e. when $u_{max} = 110 \text{ m s}^{-1}$, $\theta_{min} = 40^\circ$; [1 mark]

- Initial velocity axis asymptotic i.e. line does not meet the u -axis; [1 mark]

[Total: 4 marks]

To figure out the relationship between the variables on the axes, look for an equation linking the two.

In this case, $\sin \theta_{\min} = \frac{\sqrt{2as}}{u_{\max}}$ so θ is inversely proportional to u , giving the shape of the graph in the mark scheme.

1c

(c) Determine the length of the fuse:

List the known quantities:

- Final velocity, $v = 0$
- Angle to horizontal, $\theta = 75^\circ$
- Maximum height, $s = 250$ m
- Acceleration of free fall, $a = -g = -9.81 \text{ m s}^{-2}$

Determine the vertical component of the initial velocity:

- $v^2 = u^2 + 2as \Rightarrow u_v = \frac{\sqrt{2as}}{\sin \theta}$
- $u_v = \frac{\sqrt{2 \times 9.81 \times 250}}{\sin 75^\circ}$
- $u_v = 72.5 \text{ m s}^{-1}$ [1 mark]

Choose an appropriate kinematic equation to calculate t :

- $v = u + at \Rightarrow t = \frac{v - u}{a}$
- $t = \frac{0 - u_v}{g} = \frac{72.5}{9.81}$ [1 mark]
- $t = 7.391 = 7.4 \text{ s}$ [1 mark]

Determine the length of the fuse:

- For a burn rate of 1.5 cm s^{-1}
- Length = $1.5 \times 7.4 = 11.1 \text{ cm}$ [1 mark]

[Total: 4 marks]

1d

(d) Deduce whether the firework ignites before it lands on the hillside:

List the known quantities:

- Maximum height = s_v
- Initial velocity, $u = 75 \text{ m s}^{-1}$
- Final vertical velocity, $v = 0$
- Acceleration, $a = -g = -9.81 \text{ m s}^{-2}$
- Time until ignition, $t = 7.4 \text{ s}$ (from **(c)**)

Determine the horizontal and vertical components of initial velocity:

- $u_v = 75 \sin 30^\circ = 37.5 \text{ m s}^{-1}$
- $u_h = 75 \cos 30^\circ = 65.0 \text{ m s}^{-1}$

Determine the maximum height reached by the firework:

- $v^2 = u^2 + 2as \Rightarrow 0 = (u \sin \theta)^2 - 2gs_v$
- $s_v = \frac{(u \sin \theta)^2}{2g} = \frac{(75 \sin 30^\circ)^2}{2 \times 9.81}$
- $s_v = 71.674 = 71.7 \text{ m}$ [1 mark]

Determine the horizontal displacement of the firework when it ignites:

- $s_h = u_h t$
- $s_h = 65.0 \times 7.4 = 481 \text{ m}$ [1 mark]

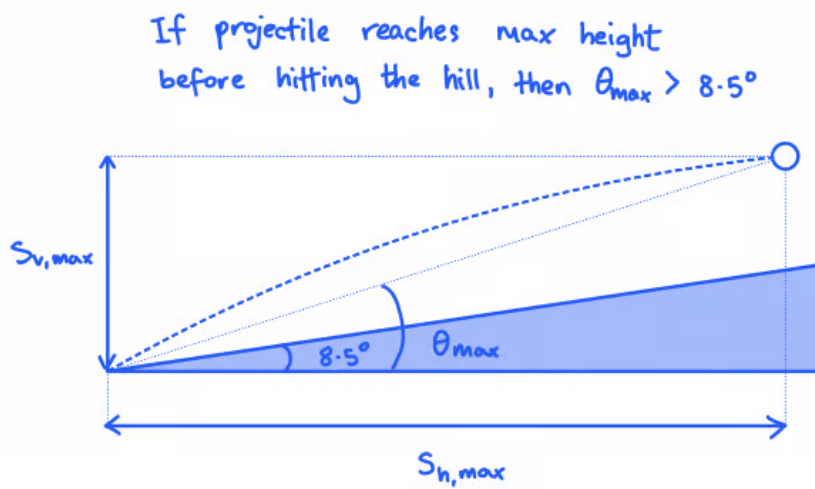
Compare the hill height 481 m away with the maximum height of the projectile:

- $h = 481 \times \tan 8.5^\circ = 71.9 \text{ m}$ [1 mark]
- $71.9 \text{ m} \approx 71.7 \text{ m}$ therefore the firework ignites just as hits the hillside; [1 mark]

[Total: 4 marks]

You could also complete this question by comparing the angle of the hill to the angle of the projectile to the horizontal

Sometimes, drawing a diagram is a good way to visualise a complicated scenario such as this one



2a

(a) Calculate the initial velocity of the motorcyclist at B:

List the known quantities:

- Resultant final velocity, $v = 13.4 \text{ m s}^{-1}$
- Angle of final velocity with horizontal, $\theta = 29.5^\circ$
- Horizontal distance travelled, $x = 6.4 \text{ m}$

Determine the vertical component of the final velocity:

- $v_v = v \sin \theta = 13.4 \sin 29.5^\circ = 6.6 \text{ m s}^{-1}$ [1 mark]

Determine the time in motion between B and C:

- In vertical motion: $v = u + at \Rightarrow v_v = gt$

- $t = \frac{v_v}{g} = \frac{6.6}{9.81} = 0.673 \text{ s}$ [1 mark]

Calculate the initial horizontal velocity at B:

- In horizontal direction: $x = u_h t$

- $u_h = \frac{x}{t} = \frac{6.4}{0.673}$ [1 mark]

- $u_h = 9.51 = 9.5 \text{ m s}^{-1}$ [1 mark]

[Total: 4 marks]

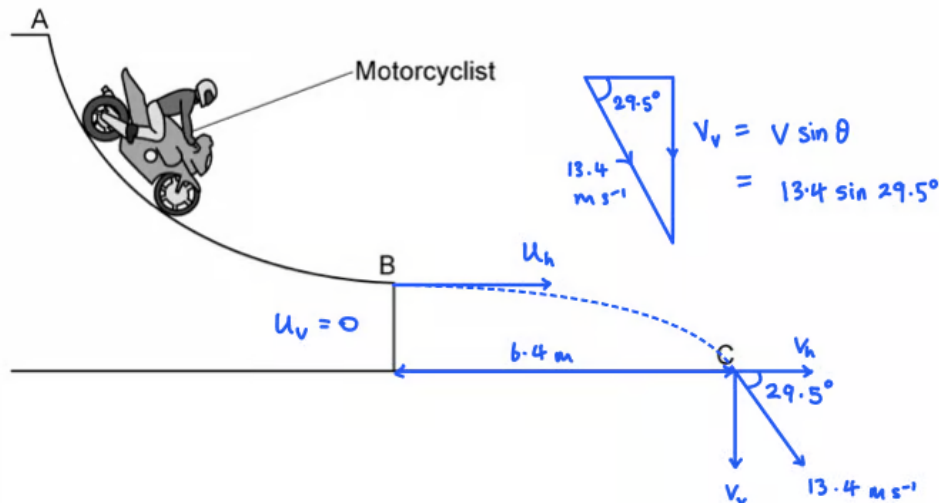
With projectile motion, you **must** split up the motion between horizontal and vertical components. Note that between B and C, the motorcyclist initially is horizontal, so it has no initial vertical velocity until it starts to move off the ramp, therefore $u_v = 0$.



- $u_h = \frac{v}{t} = \frac{6.25}{0.673}$ [1 mark]
- $u_h = 9.51 = 9.5 \text{ m s}^{-1}$ [1 mark]

[Total: 4 marks]

With projectile motion, you **must** split up the motion between horizontal and vertical components. Note that between B and C, the motorcyclist initially is horizontal, so it has no initial vertical velocity until it starts to move off the ramp, therefore $u_v = 0$.



2b

(b) Calculate the minimum initial velocity the motorcyclist requires in order to reach the other ramp:

List the known quantities:

- Angle of incline, $\theta = 25^\circ$
- Height of first ramp = 3 m
- Height of second ramp = 1 m
- Horizontal distance, $x = 24 \text{ m}$

Determine an expression for the initial velocity, u in terms of time t :

- In the horizontal direction: $x = 24 \text{ m}$, $u_h = u \cos 25^\circ$, $a = 0$, $t = ?$
- In the horizontal direction: $x = u_h t$

- $x = (u \cos 25^\circ)t \Rightarrow u = \frac{x}{t \cos 25^\circ}$
- $u = \frac{24}{t \cos 25^\circ}$ [1 mark] [Eq. 1]

Determine equation for time t from the vertical direction:

- In the vertical direction: $\Delta s = -2 \text{ m}$, $u_v = u \sin 25^\circ$, $a = -9.81 \text{ m s}^{-2}$, $t = ?$
- In the vertical direction: $s = ut + \frac{1}{2}at^2 \Rightarrow \Delta s = u_v t - \frac{1}{2}gt^2$
- $-2 = (u \sin 25^\circ)t - \left(\frac{9.81}{2}\right)t^2$ [1 mark] [Eq. 2]

Calculate t using simultaneous equations:

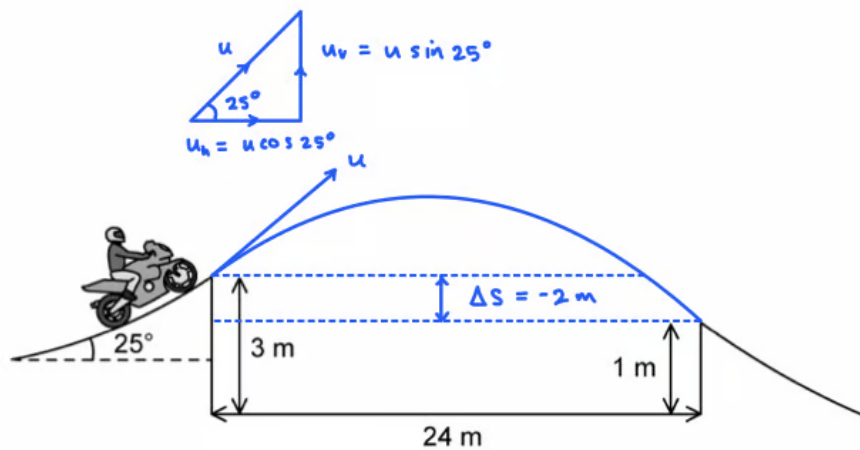
- Substitute u from [Eq. 1] into [Eq. 2]
- $-2 = \frac{24 t \sin 25^\circ}{t \cos 25^\circ} - \left(\frac{9.81}{2}\right)t^2$
- $\left(\frac{9.81}{2}\right)t^2 = 24 \tan 25^\circ + 2 \Rightarrow t^2 = \frac{48 \tan 25^\circ + 4}{9.81}$
- $t = \sqrt{\frac{48 \tan 25^\circ + 4}{9.81}} = 1.64 \text{ s}$ [1 mark]

Calculate initial velocity using [Eq. 1]:

- $u = \frac{24}{1.64 \times \cos 25^\circ} = 16 \text{ m s}^{-1}$ [1 mark]

[Total: 4 marks]

The key to this kind of problem is to consider the **total vertical displacement** due to the difference in initial and final heights



2c

(c) Deduce whether the rider lands on the other side of the river:

List the known quantities:

- Required horizontal distance, $x = 100 \text{ m}$
- Vertical height above ground, $h = 3.00 \text{ m}$
- Initial velocity, $u = 35 \text{ m s}^{-1}$
- Angle of incline of ramp, $\theta = 25^\circ$
- Horizontal acceleration $= 0$
- Vertical acceleration, $a = -g = -9.81 \text{ m s}^{-2}$

Determine the horizontal and vertical components of the initial velocity:

- Resolving horizontally ($\rightarrow$): $u_x = 35 \times \cos 25^\circ = 31.7 \text{ m s}^{-1}$
- Resolving vertically ($\uparrow$): $u_y = 35 \times \sin 25^\circ = 14.8 \text{ m s}^{-1}$

Determine the time taken to travel 100 m horizontally:

- Horizontal motion: $s = u_x t$
- $t = \frac{100}{31.7} = 3.155 \text{ s}$ [1 mark]

Determine the vertical distance fallen in this time:

- Vertical motion: $s = u_y t + \frac{1}{2} a t^2$

- $s = (14.8 \times 3.155) - \frac{1}{2} \times (9.81 \times 3.155^2) = -2.13 \text{ m}$ [1 mark]
- Vertical distance fallen = 2.1 m [1 mark]

Write a concluding statement:

- When the rider has travelled 100 m horizontally, they have fallen a distance of 2.1 m which is less than 3.0 m, hence the rider lands on the other side of the river; [1 mark]

[Total: 4 marks]

The method shown above is by far the simplest way to tackle this question and requires the fewest calculation steps with no need for a quadratic formula

There are other methods which give equally clear results but are more prone to calculation errors along the way

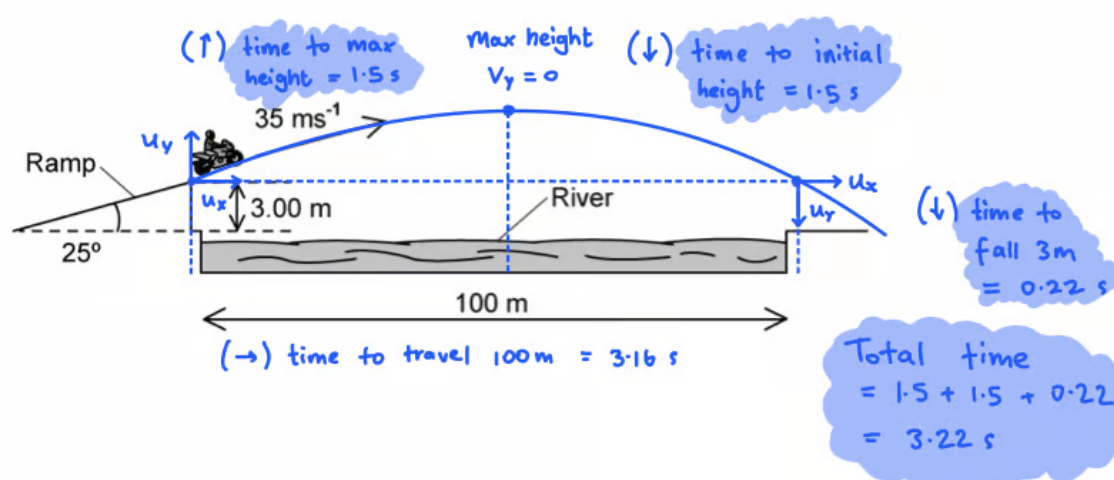
- **Alternative method 1: compare the times of flight**

Expected conclusion: it takes the rider 3.16 s to travel 100 m horizontally which is less than 3.22 s, the total time of flight, hence the rider lands on the other side of the river

- **Alternative method 2: compare the horizontal distances**

Expected conclusion: the rider travels a total horizontal distance of 102 m which is greater than 100 m, hence the rider lands on the other side of the river

It can be useful to draw a clear sketch to help yourself to visualise the situation:



2d

(d) The effect of air resistance on the rider's motion would be:

- Air resistance acts to oppose the motion of the motorcyclist; [1 mark]

Max **one** from:

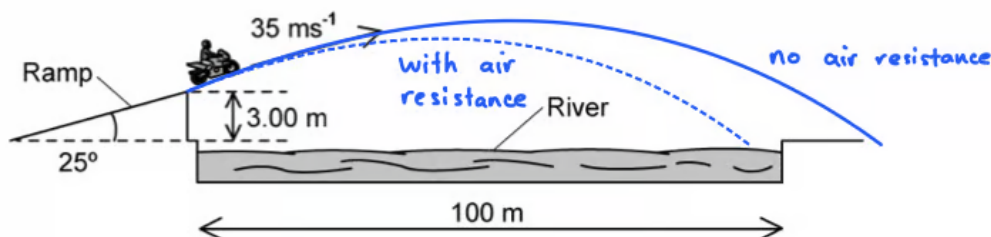
- It decreases the time for which the motorcyclist is in the air; [1 mark]
- It causes deceleration in the horizontal direction; [1 mark]
- It reduces speed in the horizontal direction; [1 mark]
- It reduces the (maximum) height reached by the motorcyclist; [1 mark]

If any mark is gained from the above, max **one** for:

- It reduces the horizontal distance travelled; [1 mark]

[Total: 3 marks]

You could also use a diagram to aid your description - a clearly annotated sketch could earn you some marks



3a

(a)

(i) The significance of the negative values of s is that its direction is:

- Upwards / away from the Moon is taken as positive **OR** downwards / towards the Moon is taken as negative; [1 mark]

(ii) An assumption about the motion of the object is:

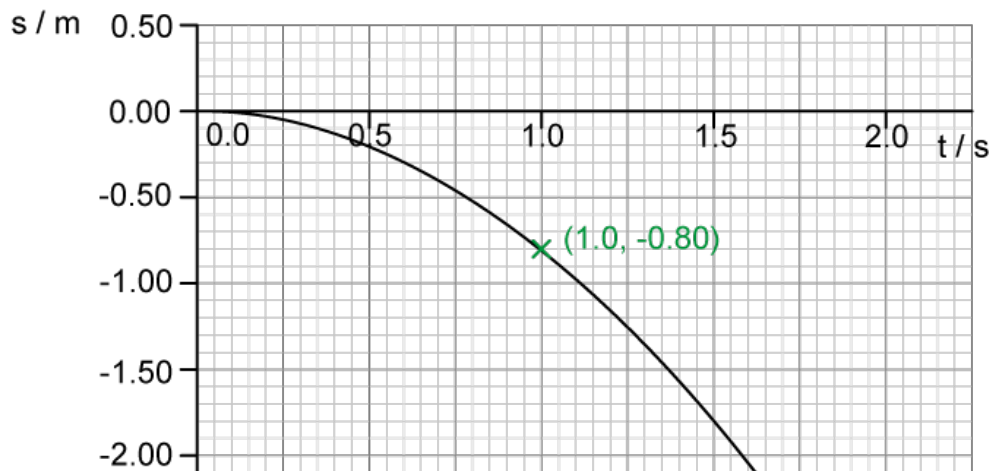
- There is no effect of air resistance; [1 mark]
- There is no gravitational effect from the Earth / another body; [1 mark]

[Total: 3 marks]

3b

(b) Use the graph to determine a value for g close to the surface of the Moon:

- Take a point on the curve:



3a

(a)

(i) The significance of the negative values of s is that its direction is:

- Upwards / away from the Moon is taken as positive **OR** downwards / towards the Moon is taken as negative; [1 mark]

(ii) An assumption about the motion of the object is:

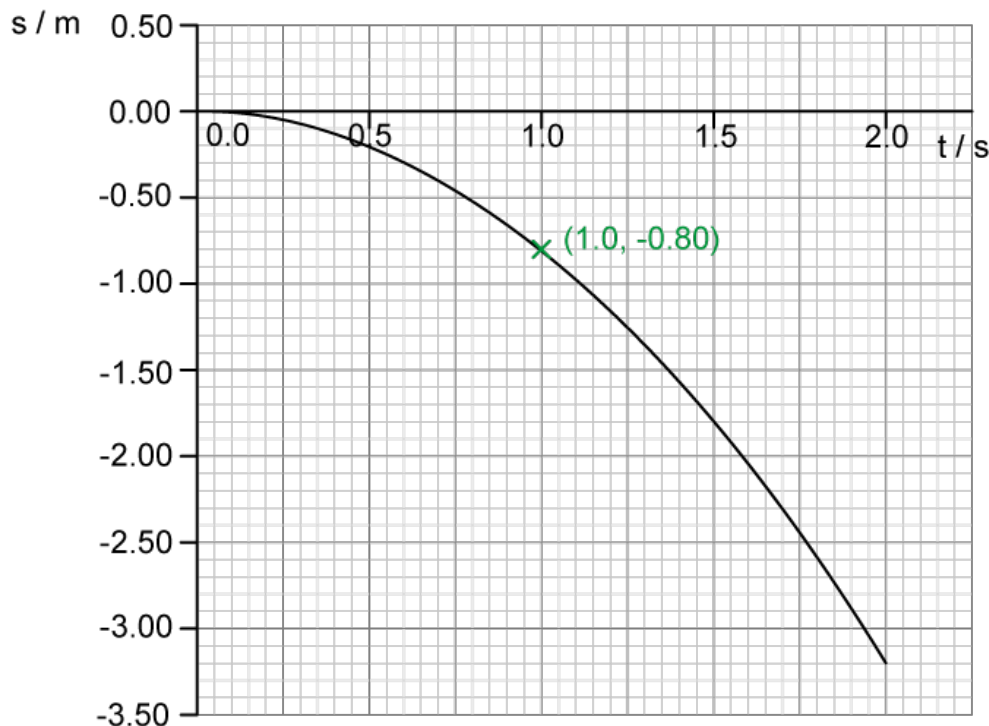
- There is no effect of air resistance; [1 mark]
- There is no gravitational effect from the Earth / another body; [1 mark]

[Total: 3 marks]

3b

(b) Use the graph to determine a value for g close to the surface of the Moon:

- Take a point on the curve:



- A suitable point labelled on the graph **OR** coordinates stated, e.g. when $t = 1.0$ s, $s = -0.80$ m; [1 mark]

Substitute coordinates into the equation of motion to find a :

- $s = ut + \frac{1}{2}at^2$
- $-0.80 = 0 + \left(\frac{1}{2} \times a \times 1.0\right)$
- $a = (2 \times -0.80) = -1.6 \text{ m s}^{-2}$ [1 mark]

[Total: 2 marks]

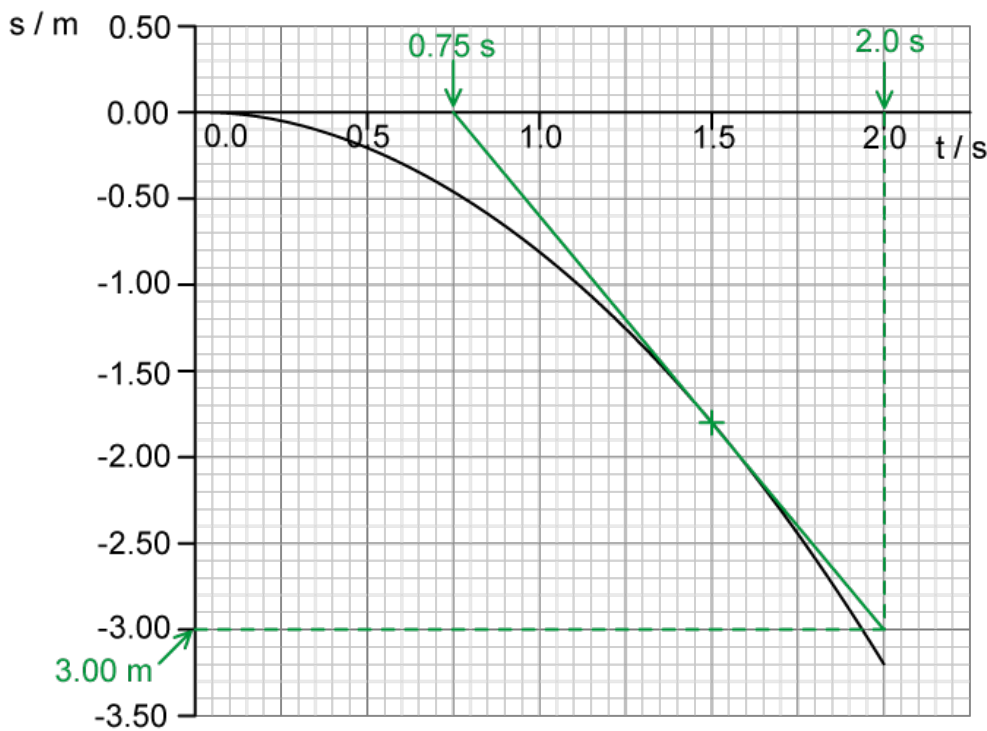
Any object accelerating uniformly is described by the equation $s = ut + \frac{1}{2}at^2$

You should realise that this is **the equation of a graph**, specifically, a displacement-time graph – as shown in this question. This is because $s \propto -t^2$ and the graph of $y \propto -x^2$ is a 'n' shape.

Hence, each coordinate on the graph satisfies this equation for s and t (where a is the acceleration due to free fall and u is the initial velocity)

3c

(c) Use the graph to estimate the instantaneous velocity:



- Tangent drawn at $t = 1.5 \text{ s}$ [1 mark]

Determine the gradient of the tangent:

- Gradient of the tangent = instantaneous velocity at that point

- $\text{gradient} = \frac{\text{change in } y}{\text{change in } x} = \frac{-3.00}{(2.0 - 0.75)}$ [1 mark]

- Instantaneous velocity: $v = -2.4 \text{ m s}^{-1}$ [1 mark]

[Total: 3 marks]

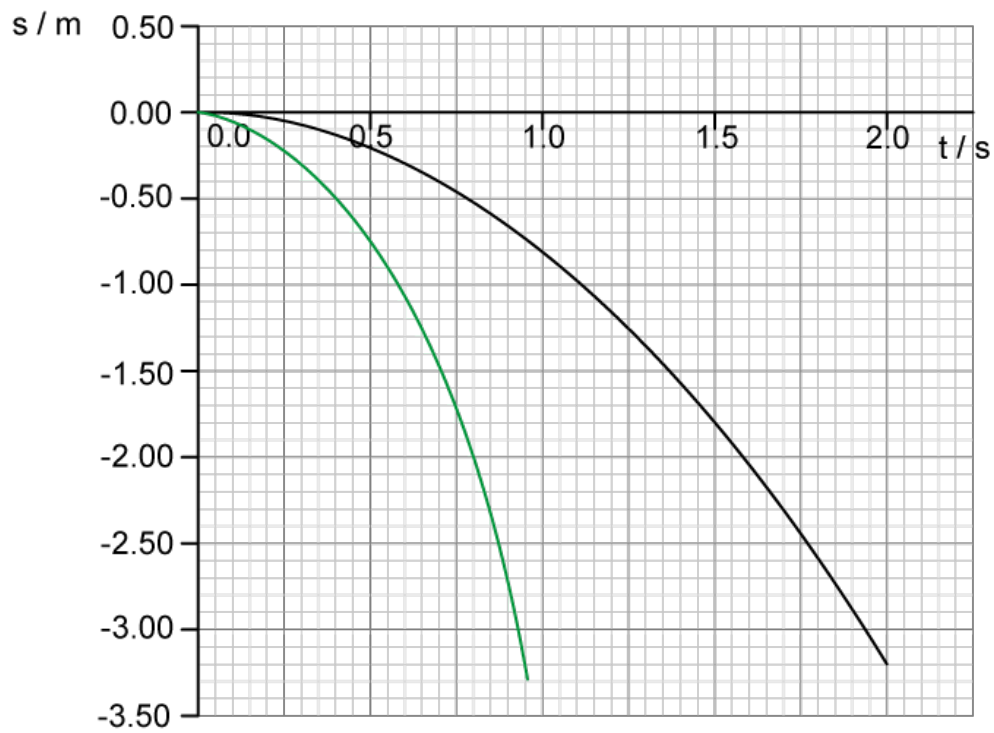
The gradient of a displacement-time graph represents the **velocity** of the object

Therefore, the gradient of the tangent at $t = 1.5 \text{ s}$ estimates the **instantaneous velocity** of the object

3d

(d)

(i) Sketch a displacement-time graph that would show the same object released close to the surface of Earth:



- A smooth parabola (judged by eye, as shown); [1 mark]
- A line always below the Moon curve (as shown); [1 mark]

- Tangent drawn at $t = 1.5 \text{ s}$ [1 mark]

Determine the gradient of the tangent:

- Gradient of the tangent = instantaneous velocity at that point

- $gradient = \frac{\text{change in } y}{\text{change in } x} = \frac{-3.00}{(2.0 - 0.75)}$ [1 mark]

- Instantaneous velocity: $v = -2.4 \text{ m s}^{-1}$ [1 mark]

[Total: 3 marks]

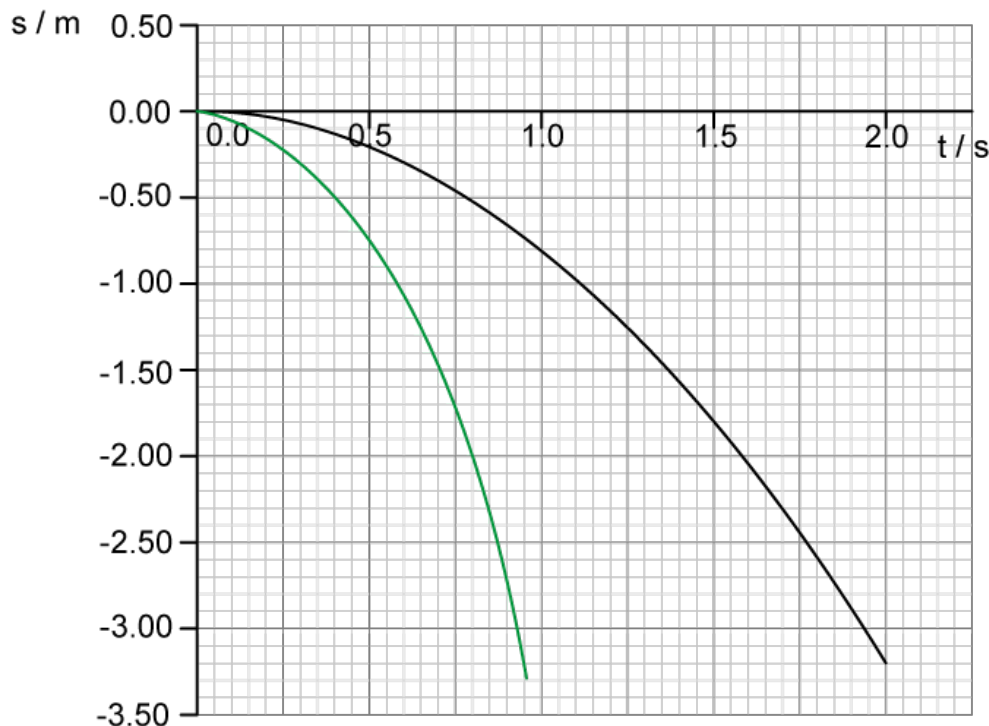
The gradient of a displacement-time graph represents the **velocity** of the object

Therefore, the gradient of the tangent at $t = 1.5 \text{ s}$ estimates the **instantaneous velocity** of the object

3d

(d)

(i) Sketch a displacement-time graph that would show the same object released close to the surface of Earth:



- A smooth parabola (judged by eye, as shown); [1 mark]
- A line always below the Moon curve (as shown); [1 mark]

(ii) Describe and explain the features of the sketch:

- The line is parabolic because the displacement s is proportional to $t^2 / s \propto t^2$ [1 mark]
- The line is below the Moon curve because the acceleration due to gravity / g on Earth is larger than on the Moon; [1 mark]

[Total: 4 marks]

If you're ever stuck on a question after part (a), make sure to look at the previous parts for clues to help you answer the question further along. From (b) we calculated the acceleration due to gravity on the moon and this is much less than the Earth. With your knowledge of $s-t$ graphs (the gradient is the velocity) then the rate of change of that gradient (the acceleration) must be bigger for the graph for Earth i.e a steeper curve.

